Name : Khushi **Date : 01/03/2026**

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LAB ASSIGNMENT – 3

Objectives :

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

DDL Questions (Data Definition Language)

1.) Create the TEACHER table with the following attributes:
(Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

```
CREATE DATABASE KHUSHI_LABASSIGN_3;
USE KHUSHI_LABASSIGN_3;
CREATE TABLE TEACHER (
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);
```

2.) Create the SALES table with the following attributes:
(Sales_No, Salesname, Branch, Salesamount, DOB).

```
CREATE TABLE SALES (
    Sales_No INT PRIMARY KEY,
    Salesname VARCHAR(50),
    Branch VARCHAR(50),
    Salesamount DECIMAL(10,2),
    DOB DATE
);
```

3.) Create the DEPARTMENTS table with the following attributes:
(dept_no, dept_name, dept_location).

```
CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
```

4.) Create the EMPLOYEES table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

```
CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

DML Questions (Data Manipulation Language)

From TEACHER Table:

1.) Insert five records into the TEACHER table.

```
INSERT INTO TEACHER VALUES
('Khushi', 1, '2018-06-15', 'Mathematics', 'Delhi', 50000),
('Neha', 2, '2019-07-20', 'Commerce', 'Mumbai', 45000),
('Rashi', 1, '2017-03-12', 'Mathematics', 'Delhi', 55000),
('Simran', 3, '2020-01-10', 'Physics', 'Chennai', 40000),
('Aayra', 2, '2016-09-05', 'Commerce', 'Mumbai', 48000);
SELECT* FROM TEACHER ;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Khushi	1	2018-06-15	Mathematics	Delhi	50000.00
Neha	2	2019-07-20	Commerce	Mumbai	45000.00
Rashi	1	2017-03-12	Mathematics	Delhi	55000.00
Simran	3	2020-01-10	Physics	Chennai	40000.00
Aayra	2	2016-09-05	Commerce	Mumbai	48000.00

2.) Give an increment of 25% salary to teachers working in the Mathematics department.

```
SET SQL_SAFE_UPDATES = 0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
SELECT* FROM TEACHER;
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Khushi	1	2018-06-15	Mathematics	Delhi	62500.00
Neha	2	2019-07-20	Commerce	Mumbai	45000.00
Rashi	1	2017-03-12	Mathematics	Delhi	68750.00
Simran	3	2020-01-10	Physics	Chennai	40000.00
Aayra	2	2016-09-05	Commerce	Mumbai	48000.00

3.) Give an increment of 15% salary to teachers working in the Commerce department.

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
SELECT* FROM TEACHER;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell C

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Khushi	1	2018-06-15	Mathematics	Delhi	62500.00
Neha	2	2019-07-20	Commerce	Mumbai	51750.00
Rashi	1	2017-03-12	Mathematics	Delhi	68750.00
Simran	3	2020-01-10	Physics	Chennai	40000.00
Aayra	2	2016-09-05	Commerce	Mumbai	55200.00

From SALES Table

1.) Insert five records into the SALES table.

```
INSERT INTO SALES VALUES
(101, 'Rashi', 'North', 20000, '1998-12-21'),
(102, 'Priya', 'South', 30000, '1997-05-10'),
(103, 'Khushi', 'East', 25000, '1996-12-11'),
(104, 'Meenal', 'West', 40000, '1995-08-15'),
(105, 'Vanshika', 'North', 15000, '1999-12-05');
SELECT* FROM SALES;
```

OUTPUT :

Result Grid | Filter Rows: | Edit:

Sales_No	Salesname	Branch	Salesamount	DOB
101	Rashi	North	20000.00	1998-12-21
102	Priya	South	30000.00	1997-05-10
103	Khushi	East	25000.00	1996-12-11
104	Meenal	West	40000.00	1995-08-15
105	Vanshika	North	15000.00	1999-12-05
NULL	NULL	NULL	NULL	NULL

2.) Calculate the total sales amount in each branch.

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT :

Result Grid | Filter

Branch	Total_Sales
North	35000.00
South	30000.00
East	25000.00
West	40000.00

3.) Calculate the average sales amount in each branch.

```
SELECT Branch, AVG(Salesamount) AS Avg_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT :

Result Grid | Filter Ro

Branch	Avg_Sales
North	17500.000000
South	30000.000000
East	25000.000000
West	40000.000000

4.) Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.

```
SELECT Salesname,
DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

OUTPUT:

Result Grid | Filter Row

Salesname	DOB
Rashi	21-Dec-98
Khushi	11-Dec-96
Vanshika	05-Dec-99

5.) Display the Salesname and DOB of salesmen in alphabetical order of the month.

```
SELECT Salesname, DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

OUTPUT:

Result Grid | Filter Rou

Salesname	DOB
Meenal	1995-08-15
Rashi	1998-12-21
Khushi	1996-12-11
Vanshika	1999-12-05
Priya	1997-05-10

From JOIN Operations:

First We Insert Records for JOIN:

Insert into DEPARTMENTS

```
INSERT INTO DEPARTMENTS VALUES
(1, 'Mathematics', 'Delhi'),
(2, 'Commerce', 'Mumbai'),
(3, 'Physics', 'Gurugram');
SELECT* FROM DEPARTMENTS;
```

OUTPUT :

Result Grid | Filter Rows:

dept_no	dept_name	dept_location
1	Mathematics	Delhi
2	Commerce	Mumbai
3	Physics	Gurugram
NULL	NULL	NULL

Insert into EMPLOYEES

```
INSERT INTO EMPLOYEES VALUES
(1, 'Ankita', 50000, 1),
(2, 'Bhavna', 45000, 2),
(3, 'Khushi', 60000, 1),
(4, 'Disha', 35000, 3),
(5, 'Ekta', 70000, 2);
SELECT* FROM EMPLOYEES;
```

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Ankita	50000.00	1
2	Bhavna	45000.00	2
3	Khushi	60000.00	1
4	Disha	35000.00	3
5	Ekta	70000.00	2
NULL	NULL	NULL	NULL

1.) Display employee details and department details where the department numbers are the same in both tables.

```
SELECT *
FROM EMPLOYEES E
INNER JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
1	Ankita	50000.00	1	1	Mathematics	Delhi
3	Khushi	60000.00	1	1	Mathematics	Delhi
2	Bhavna	45000.00	2	2	Commerce	Mumbai
5	Ekta	70000.00	2	2	Commerce	Mumbai
4	Disha	35000.00	3	3	Physics	Gurugram

2.) Display the employee name and department name using a LEFT OUTER JOIN.

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT:

Result Grid | Filter Rows

emp_name	dept_name
Ankita	Mathematics
Bhavna	Commerce
Khushi	Mathematics
Disha	Physics
Ekta	Commerce

3.) Display the employee name and department name using a RIGHT OUTER JOIN.

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT :

Result Grid | Filter Rows

emp_name	dept_name
Ankita	Mathematics
Khushi	Mathematics
Bhavna	Commerce
Ekta	Commerce
Disha	Physics

4.) Display the details of employees who draw a salary greater than the average salary.

```
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

OUTPUT :

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
3	Khushi	60000.00	1
5	Ekta	70000.00	2
NULL	NULL	NULL	NULL

TCL Questions (Transaction Control Language)

1.) Perform the ROLLBACK command on the TEACHER table.

```
SET autocommit = 0;
UPDATE TEACHER
SET Salary = Salary + 1000;
ROLLBACK;
SELECT * FROM TEACHER;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Khushi	1	2018-06-15	Mathematics	Delhi	62500.00
Neha	2	2019-07-20	Commerce	Mumbai	51750.00
Rashi	1	2017-03-12	Mathematics	Delhi	68750.00
Simran	3	2020-01-10	Physics	Chennai	40000.00
Aayra	2	2016-09-05	Commerce	Mumbai	55200.00

2.) Perform the COMMIT command on the TEACHER table.

```
UPDATE TEACHER
SET Salary = Salary + 1000;
COMMIT;
SELECT* FROM TEACHER;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell C

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Khushi	1	2018-06-15	Mathematics	Delhi	63500.00
Neha	2	2019-07-20	Commerce	Mumbai	52750.00
Rashi	1	2017-03-12	Mathematics	Delhi	69750.00
Simran	3	2020-01-10	Physics	Chennai	41000.00
Aayra	2	2016-09-05	Commerce	Mumbai	56200.00

3.) Write a query to implement a SAVEPOINT.

```
UPDATE TEACHER SET Salary = Salary + 100;
SAVEPOINT A;
SELECT* FROM TEACHER;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Khushi	1	2018-06-15	Mathematics	Delhi	63800.00
Neha	2	2019-07-20	Commerce	Mumbai	53050.00
Rashi	1	2017-03-12	Mathematics	Delhi	70050.00
Simran	3	2020-01-10	Physics	Chennai	41300.00
Aayra	2	2016-09-05	Commerce	Mumbai	56500.00

```
UPDATE TEACHER SET Salary = Salary + 1000;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Khushi	1	2018-06-15	Mathematics	Delhi	64800.00
Neha	2	2019-07-20	Commerce	Mumbai	54050.00
Rashi	1	2017-03-12	Mathematics	Delhi	71050.00
Simran	3	2020-01-10	Physics	Chennai	42300.00
Aayra	2	2016-09-05	Commerce	Mumbai	57500.00

```
ROLLBACK TO A;
SELECT* FROM TEACHER;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell C

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Khushi	1	2018-06-15	Mathematics	Delhi	63800.00
Neha	2	2019-07-20	Commerce	Mumbai	53050.00
Rashi	1	2017-03-12	Mathematics	Delhi	70050.00
Simran	3	2020-01-10	Physics	Chennai	41300.00
Aayra	2	2016-09-05	Commerce	Mumbai	56500.00

DCL Questions (Data Control Language):

1.) Write a query to grant all privileges on the EMPLOYEES table to DEPARTMENTS.

FIRST WE HAVE TO CREATE A USER.

```
CREATE USER 'departments_user'@'localhost'
IDENTIFIED BY '1234';
```

THEN WE GRANT ALL PRIVILEGES ON THE EMPLOYEE TABLE TO DEPARTMENTS .

```
GRANT ALL PRIVILEGES
ON KHUSHI_LABASSIGN_3.EMPLOYEES
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

Result Grid | Filter Rows:

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...
	GRANT ALL PRIVILEGES ON `khushi_labassign_...

2.) Write a query to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, INSERT, UPDATE
ON KHUSHI_LABASSIGN_3.EMPLOYEES
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT:

Result Grid | Filter Rows:

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...
	GRANT ALL PRIVILEGES ON `khushi_labassign_...

3.) Write a query to revoke all privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE ALL PRIVILEGES
ON KHUSHI_LABASSIGN_3.EMPLOYEES
FROM 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT:

Result Grid | Filter Rows:

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...

4.) Write a query to revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

First, we have to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, INSERT, UPDATE
ON KHUSHI_LABASSIGN_3.EMPLOYEES
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

Result Grid | Filter Rows:

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...
	GRANT SELECT, INSERT, UPDATE ON `khushi_l...

Now, we revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE SELECT, INSERT
ON KHUSHI_LABASSIGN_3.EMPLOYEES
FROM 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

OUTPUT :

Result Grid | Filter Rows:

	Grants for departments_user@localhost
▶	GRANT USAGE ON *.* TO `departments_user`...
	GRANT UPDATE ON `khushi_labassign_3`.`em...